

Total No. of Questions : 8]

SEAT No. :

P416

[Total No. of Pages : 2

[6003]-504

T.E. (Electronics & Computer Engineering)
COMPUTER NETWORKS AND SECURITY
(2019 Pattern) (Semester-II) (310353)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Black figures to the right indicate full marks.
- 3) Neat diagrams must be draw wherever necessary.

- Q1)** a) Explain the purpose of flow control and error control in data link layer. **[4]**
b) Explain the working HDLC protocol with neat diagram. **[6]**
c) What are the noisy channcls? Explain stop and wait protocol with suitable diagram. **[6]**

OR

- Q2)** a) Explain frame format of point to point protocol. **[4]**
b) What are the types of media access control? Explain in detail random access technique. **[6]**
c) Explain router with suitable diagram. **[6]**

- Q3)** a) Write a short note on Address Resolution protocol (ARP) **[6]**
b) Explain IPv6 address in detail **[6]**
c) Explain Internet control message protocol (ICMP) with suitable architecture diagram. **[6]**

OR

- Q4)** a) Explain Transmission Control Protocol with header diagram. **[6]**
b) Explain in detail link state routing vector with suitable diagram. **[6]**
c) What is congestion control? List the typical QoS parameters in the transport layer and explain each one. **[6]**

P.T.O.

- Q5)** a) Explain with architecture of WWW. [4]
b) What is DNS? Explain Domain Name Space in detail. [6]
c) What are the different commands used in SMTP. Explain Simple Mail Transfer protocol. [8]

OR

- Q6)** a) What is streaming of Audio/Video. Explain. [4]
b) Write a short note on Email with message format. [6]
c) Explain SNMP with suitable diagram. What is the role of structure management Information in SNMP explain. [8]

- Q7)** a) List the different security goals. Explain. [4]
b) Explain public key cryptography (cipher) with suitable diagram. [6]
c) What is Network Monitoring? How to perform network monitoring effectively. [8]

OR

- Q8)** a) Write a short note on network tester. [4]
b) Explain with neat diagram internet access through dial up. [6]
c) What is Protocol Analyzer? Explain the working with neat diagram. [8]



Total No. of Questions : 8]

SEAT No. :

P7612

[Total No. of Pages : 2

[6180]-131

T.E. (Electronics & Computer Engineering)
COMPUTER NETWORKS AND SECURITY
(2019 Pattern) (Semester-II) (3103353)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Neat diagrams must be drawn wherever necessary.*
- 2) *Figures to the right indicate full marks.*

- Q1)** a) List and Explain functions of data link layer. [4]
- b) What is need of framing? What are the different technique of framing? Explain any two. [6]
- c)** What are the noiseless channels? Explain simplest protocol with suitable diagram. [6]

OR

- Q2)** a) What is point to point protocol? Explain in detail [4]
- b) What are the types of media access control? Explain in detail controlled access technique. [6]
- c) Explain Hub with suitable diagram. [6]
- Q3)** a) Explain network layer services in detail. [6]
- b) Explain IPv4 address in detail [6]
- c) Explain Internet Group Message Protocol (IGMP) with suitable architecture diagram. [6]

OR

P.T.O.

- Q4) a)** What is congestion control? List the typical QOS parameters in the transport layer and explain each one. [6]
- b)** Explain in detail path vector routing with suitable diagram. [6]
- c)** Explain User Datagram Protocol with special features [6]

- Q5) a)** Explain feature of HTTP. [4]
- b)** Explain TELNET in detail with respect to server and client communication.[6]
- c)** What the different commands used in FTP? Explain File transfer Protocol in detail. [8]

OR

- Q6) a)** What is streaming of Audio / Video. Explain. [4]
- b)** What is DNS? Explain Domain Name Space in detail. [6]
- c)** Explain SNMP with suitable diagram. What is the role of structure management Information in SNMP explain. [8]

- Q7) a)** List the different security goals. Explain. [4]
- b)** Explain secret key cryptography (cipher) with suitable diagram. [6]
- c)** What is Network Monitoring? How to perform network monitoring effectively [8]

OR

- Q8) a)** Explain UTP cabling for PC to PC [4]
- b)** Explain with neat diagram internet access through leased line. [6]
- c)** What is Protocol Analyzer? Explain the working with neat diagram [8]



Total No. of Questions : 8]

SEAT No. :

PC1795

[6353]-114

[Total No. of Pages :2

T.E. (Electronics and Computer Engineering)
COMPUTER NETWORKS AND SECURITY
(2019 Pattern) (Semester - II) (310353)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume Suitable data if necessary.

- Q1)** a) Explain Random Access Techniques in detail. [8]
b) Describe in brief HDLC frame format. [6]
c) Differentiate between Noisy channel and Noiseless channel. [4]

OR

- Q2)** a) Explain Services provided by the data link layer. [8]
b) Explain CSMA/CD Protocol. [6]
c) Write a note on Backbone networks [4]

- Q3)** a) What is use of IP address? Explain in brief IPV4 classful addressing technique. [7]
b) Write a note on Address Resolution Protocol. [6]
c) Differentiate between Intra domain and Inter domain Routing. [4]

OR

- Q4)** a) What is mean by congestion and quality of service in computer networks. [7]
b) Explain shortest path algorithm. [6]
c) Compare TCP and UDP. [4]

P.T.O.

- Q5)** a) List out the different commands used in SMTP? Explain Simple mail transfer protocol. [8]
- b) What is HTML? Explain with any application. [6]
- c) Explain working of Ping and Trace route commands. [4]

OR

- Q6)** a) Explain SNMP with suitable diagram. Explain role of structure management information in SNMP. [8]
- b) Write a note on. [6]
- i) Email
- ii) WWW
- c) Explain Telnet with example. [4]

- Q7)** a) What is public key cryptography? Explain RSA algorithm in detail. [7]
- b) Write a note on Hash function. [6]
- c) Write a note on Network tester. [4]

OR

- Q8)** a) What is network monitoring. how to perform network monitoring effectively. [7]
- b) Explain Basics of security requirement. [6]
- c) Explain working of DSL. [4]



Total No. of Questions : 8]

SEAT No. :

PB3853

[Total No. of Pages : 2

[6262]-115

T.E. (Electronics & Computer Engineering)

COMPUTER NETWORKS & SECURITY

(2019 Pattern) (Semester - II) (310353)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Figures to the right indicate full marks.
- 3) Neat diagrams must be drawn wherever necessary.

- Q1)** a) What are noiseless and noisy channels? [4]
- b) Explain CYCLIC CODES? Explain CRC code with example. [6]
- c) Explain Multiple access techniques-Channelization in detail. [6]

OR

- Q2)** a) Write difference between Hub and Bridge. [4]
- b) Explain POINT- TO-POINT PROTOCOL (PPP)? Which services Provided by PPP. [6]
- c) Explain following Ethernet types (Any 2)- [6]
- i) Bridged,
 - ii) Switched,
 - iii) Full duplex

- Q3)** a) Short note (Any 1) [6]
- i) RARP
 - ii) IPv4 datagram
- b) Explain working of DHCP with suitable diagram. [6]
- c) Explain architecture of Internet Control Message Protocol (ICMP). [6]

OR

P.T.O.

- Q4) a) Short note (Any 1) [6]**
- i) ARP
 - ii) IGMP
- b) Explain Routing? Explain different types of Routing Algorithms. [6]
- c) Compare Connection-Oriented and Connectionless Service with respect to Data Reliability, Congestion, Bandwidth Requirement, Authentication with example. [6]

- Q5) a) Explain File Transfer Protocol (FTP) in Application Layer. [4]**
- b) What is the socket () function in socket programming? Explain elements of socket? [6]
- c) Write short note on any two of following Application Layer Protocols: DNS, FTP, SMTP, And MIME Protocols. [8]

OR

- Q6) a) Explain any 4 Features of MIME. [4]**
- b) Explain Domain Name System with suitable diagram. [6]
- c) Which are different commands used in SMTP? Explain. [8]

- Q7) a) What are secret key algorithms public key algorithms? [4]**
- b) What are the 3 main types of cryptographic algorithms? [6]
- c) What are the 4 types of attacks in network security? [8]

OR

- Q8) a) Write Features of Cryptography in detail. [4]**
- b) Compare Encryption and Cryptography. [6]
- c) Which tools is used to allow simulating networks? List any 4 tools. [8]

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